

Enhancement Three Narrative

1. Briefly describe the artifact. What is it? When was it created?

The artifact is the same project I have been working on. For this enhancement the main artifacts are the database files such as schema.prisma and seed.js, though multiple other files have been modified to accommodate the database changes.

Here are the modified files:

Backend:

- schema.prisma – removed Home model and added fields to the Community model
- seed.js – added AI generated script that creates new communities for development and test. Total 60 communities now in the test database
- app.js – removed the routes for homes
- communityController.js – removed all references to homes, added home fields to community methods
- homeController.js – deleted (not needed anymore)
- homes.js – deleted (not needed anymore)

Frontend:

- App.jsx – removed methods not needed for homes and updated community methods. Added method to center the map on the selected community
- AddCommunityModal.jsx – updated the modal for users to add communities so the fields previous in homes is now in communities
- AddCommunityModal.css – updated css for user add community modal changes
- CommunityCard.jsx – updated the popup card for the community changes
- MapView.jsx – changes for removing the home fields. Also added support for the recenter focus on selected community feature

2. Justify the inclusion of the artifact in your ePortfolio. Why did you select this item? What specific components of the artifact showcase your skills and abilities in software development? How was the artifact improved?

For the Databases enhancement, I made some enhancements to the database and corresponding changes throughout the project. The biggest change was removing the Home modal that the generative AI created. I want this web application to focus strictly on communities. Many of the fields from Home had to be tweaked and added to the Community modal, but I think the overall structure is cleaner for users and future

developers. This required multiple updates to the controllers and the frontend UI. Another major change was adding a total of 60 communities to the database for development. I again used AI to help generate these example communities for testing. This kind of tedious repetitive work, especially for testing, is perfect use of AI tools.

3. Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

The focus of my third enhancement is the database itself. The database schema created by the generative AI was not working as I intended, so I planned these updates. There are two outcomes I wanted to work on for this enhancement. First, “Employ strategies for building collaborative environments...” I planned to meet by making the database more developer friendly. Instead of multiple modals that are meshed together in the end, I put all the fields in one modal. I think it is easier to understand and to maintain. Another way this enhancement supports this outcome is by creating the seed.js that provides example communities to test the application. I am developing and testing my own work in this class, but in a software team normally other engineers would test the application. This kind of test data built into the application is invaluable for testing, not just the database, but also the frontend UI and all the API routing.

The second outcome planned for this enhancement was “Demonstrate an ability to use well-founded and innovative techniques...”. I worked towards this outcome by again using the generative AI to create the seed script instead of manually creating dozens of example communities. Using these tools for testing is efficient and delivers value to the customer.

Insights after finishing the project:

Collaborating in a team environment – added comments throughout to make the code easier to understand and maintain. Seeding the database makes it easier for teams to test other functions of the application. Combining models into one makes it easier to maintain the database as it grows.

Communicating with stakeholders – other home websites are bloated with information. By removing the Home model and adding simple, filterable ranges to the Community model, the overall experience is simplified for customers.

Data Structures and algorithms – adding 60 communities to the database enabled testing of the many filter and sort options and their underlying data structures and algorithms.

Software engineering and database – removed the Home model and added the necessary information to the Community model, simplifying the database structure. Had to make many updates throughout the application to accommodate the database change.

4. Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

This enhancement seemed straightforward. I just had to refactor the database schema. However, the challenging part that was semi-unexpected was seeing how many changes were necessary throughout the rest of the files to accommodate the database changes. I knew there would be some, but so much of the app interacts with the data from the database in some way. Another challenge was after I seeded the database with a lot more example communities it exposed some issues with the UI. For instance, one issue I was unaware of and had to fix was that the map was not centering on the selected community. This only became obvious once I added example communities in multiple states. I feel like the UI/UX is not quite right, but now that I have a lot more to work with, I will test it out and go through some user work flows when I polish the enhancements.